

D.A.V. LOHAGHAT
PRE-BOARD EXAMINATION-2024-25

SUBJECT - PHYSICS

MM: 70

CLASS-XII

Time: 3:00 hours

General Instructions :

Read the following instructions very carefully and strictly follow them :

1. This question paper contains **33** questions. **All** questions are **compulsory**.
2. This question paper is divided into **five** Sections **A, B, C, D** and **E**.
3. In **Section A** Questions no. **1** to **16** are Multiple Choice (MCQ) type questions, carrying **1** mark each.
4. In **Section B** Questions no. **17** to **21** are Very Short Answer (VSA) type questions, carrying **2** marks each.
5. In **Section C** Questions no. **22** to **28** are Short Answer (SA) type questions, carrying **3** marks each.
6. In **Section D** Questions no. **29** to **31** are Long Answer (LA) type questions carrying **5** marks each.
7. In **Section E** Questions no. **32** and **33** are case-based questions carrying **4** marks each.
8. There is no overall choice. However, an internal choice has been provided in 1 questions in Section B, 1 questions in Section C, 3 questions in Section D and one question in each CBQ in Section E. You have to attempt only one of the choices in such questions.
9. Use of calculators is **not** allowed.
10. Use the following values of physical constants, if required :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

$$\text{Mass of electron } (m_e) = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

Section A - MCQ

Q 1. In a Whetstone's bridge, all the four arms have equal resistance R . If resistance of the galvanometer arm is also R , then equivalent resistance of the combination is- [1]

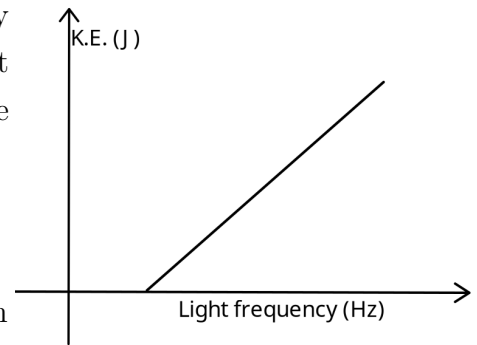
- A. $\frac{R}{4}$ B. R C. $2R$ D. $\frac{R}{2}$

Q 2. The focal lengths of the objective and the eyepiece of a compound microscope are 1 cm and 2 cm respectively. If the tube length of the microscope is 10 cm, the magnification obtained by the microscope for most suitable viewing by relaxed eye is : [1]

- A. 250 B. 200 C. 150 D. 125

Q 3. The graph shows the variation of the maximum kinetic energy of the emitted photoelectron with the frequency of the incident radiation for a given metal. Which of the following gives the work function of the metal? [1]

- A. x-intercept C. the slope of the graph
B. y-intercept D. the area under the graph



Q 4. When current in a coil changes from 5 A to 2 A in 0.1 s, average voltage of 50 V is produced. The self-inductance of the coil will be. [1]

- A. 1.67 H B. 6 H C. 3 H D. 0.67 H

Q 5. The electromagnetic radiation used for water purification and eye surgery is [1]

- A. Infrared B. Microwave C. X-rays D. Ultraviolet wave

Q 6. Three students construct a solenoid of length 35 cm. They are each given insulated copper wire of the same length. The table below lists some details about the solenoids made by them. [1]

	Magnetic field produced	Radius of solenoid	Core of solenoid
Student 1	B_1	3 cm	air
Student 2	B_2	3 cm	iron
Student 3	B_3	6 cm	air

Compare the magnetic field produced by the solenoids made by the three students.

- A. $B_1 = B_3 < B_2$ B. $B_3 < B_1 < B_2$ C. $B_1 < B_2 < B_3$ D. $B_1 = B_2 > B_3$

Q 7. A hot-wire ammeter reads 10 A in an AC circuit. The peak value of the current is [1]

- A. 5π A B. $10\sqrt{2}$ A C. $\frac{10}{\sqrt{2}}$ A D. $\frac{1}{\sqrt{2}}$ A

Q 8. If a wire of length 2 m is moving with a velocity of 1 m/s perpendicular to a magnetic field of 0.5 T, then E.M.F. induced in the wire will be [1]

- A. 0.2 V B. 1.0 V C. 0.5 V D. 2.0 V

Q 9. Two spherical nuclei have mass number 216 and 64 with their radii R_1 and R_2 respectively. The ratio $\frac{R_1}{R_2}$ is equal to [1]

- A. 1 : 3 B. 3 : 1 C. 2 : 3 D. 3 : 2

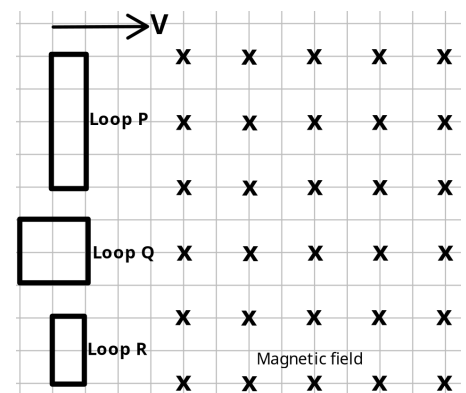
Q 10. Find the ratio of intensities at two points on a screen in Young's double slit experiment, when waves from the two slits have path difference of 0 and $\frac{\lambda}{4}$ [1]

- A. 2 : 1 B. 1 : 2 C. 2 : 3 D. 4 : 1

Q 11. Three loops as shown in the figure move into the magnetic field with a velocity v . [1]

In which loop(s) will the induced emf be the **largest** at the instant when the loops enter the magnetic field?

- A. Only P C. Only P and Q
B. Only Q D. Only Q and R



Q 12. If the orbital radius of the electron in a hydrogen atom is 47×10^{-11} m, compute the potential energy of the electron in hydrogen atom. [1]

- A. 15.3 eV B. -30.6 eV C. 13.6 eV D. -13.6 eV

Questions number **13** to **16** are Assertion (A) and Reason (R) type questions. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false and Reason (R) is also false.

Q 13. Assertion (A) : The focal length of a lens for red light is more than that of blue light ($f_r > f_b$), [1]

Reason (R) : The refractive index of material is depend on wave length of light i.e. ($\mu_b > \mu_r$)

Q 14. Assertion (A) : The depletion layer in p-n junction under forward bias decreases. [1]

Reason (R) : The electric field due to external voltage supports the electric field due to potential barrier.

Q 15. Assertion (A) : A very intense beam of light having frequency less than the threshold frequency does not cause photoelectric emission [1]

Reason (R) : Photoelectric emission is only possible if the frequency of incident radiation is greater than the threshold frequency.

Q 16. Assertion (A) : The energy of a charged particle moving in a magnetic field does not change. [1]

Reason (R) : It is because the work done by the magnetic force on the charge moving in a magnetic field is zero.

Section B - Very Short Answer Questions

Q 17. Draw binding energy curve showing the region of fission, fusion and maximum stability. [2]

Q 18. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate the magnifying power in normal adjustment and the distance of the image formed by the objective. [2]

Q 19. (a) Give construction and working of full wave rectifier. Draw input and output wave form. [2]

OR

(b) Explain, with the help of a suitable diagram, how (i) depletion layer and (ii) potential barrier is formed in a p-n junction diode. [2]

Q 20. A proton and an alpha particle both are projected in uniform magnetic field B perpendicular to their velocity with same kinetic energy. Find the ratio of their radii of circular path. [2]

Q 21. (a) The refractive index of a material of a concave lens is μ_1 . It is immersed in a medium of refractive index μ_2 . A parallel beam of light is incident on the lens. Trace the path of emergent rays when [2]

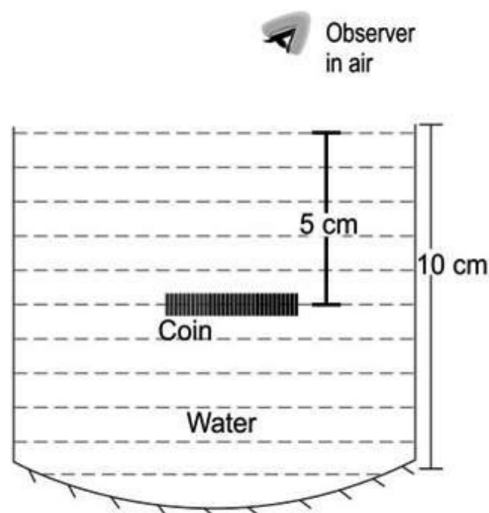
i. $\mu_2 > \mu_1$

ii. $\mu_2 < \mu_1$

OR

(b) A glass beaker of height 10 cm, completely filled with water (refractive index = $4/3$), has a curved bottom which is silvered as shown below.

A plastic coin remains submerged in water at a depth of 5 cm from the top of the beaker. An observer sees the coin in the water and its image in the mirror. If the image formed by the curved mirror is seen by the observer at a distance of 15 cm from the surface of the water, what is the focal length of the curved surface? (Assume the silvered curved surface acts as a spherical mirror.)



Section C - Short Answer Questions

- Q 22.** (a) Define self inductance and write its SI unit. Obtain an expression of self inductance of a solenoid of length 'l' cross-sectional area 'A' having 'N' no. of turns. [3]

OR

- (b) i. According to Lenz's Law. In a closed circuit, the induced current opposes the change in magnetic flux that produced it as per the law of conservation of energy. Justify. [3]
- ii. A metal rod of length 2 m is rotated with a frequency 60 rev/s about an axis passing through its centre and perpendicular to its length. A uniform magnetic field of 2 T perpendicular to its plane of rotation is switched-on in the region. Calculate the e.m.f. induced between the centre and the end of the rod.

- Q 23.** (a) Write two main points of Huygen's theory. A plane light wave propagating from a rarer into a denser medium, is incident at an angle i on the surface separating two media. Using Huygen's principle, draw the refracted wave and hence verify Snell's law of refraction. [3]

OR

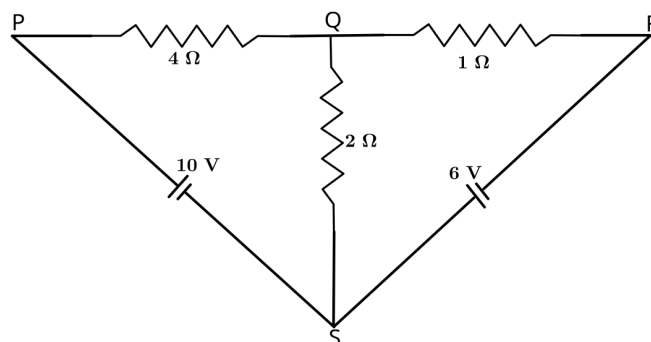
- (b) A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes in a Young's double-slit experiment. [3]
- i. Find the distance of the third bright fringe on the screen from the central maximum for wavelength 650 nm.
- ii. What is the least distance from the central maximum where the bright fringes due to both the wavelengths coincide?

- Q 24.** (a) i. Two energy levels of an electron in hydrogen atom are separated by 2.55 eV. Find the wavelength of radiation emitted when the electron makes transition from the higher energy level to the lower energy level. [3]
- ii. In which series of hydrogen spectrum this line shall fall?

OR

- (b) The earth revolves around the sun in an orbit of radius 1.5×10^{11} m with orbital speed 30 km/s. Find the quantum number that characterises its revolution using Bohr's model in this case (mass of earth = 6.0×10^{24} kg) [3]

- Q 25.** Using Kirchhoff's laws, calculate the current flowing through $4\ \Omega$, $1\ \Omega$ and $2\ \Omega$ resistors in the circuit shown below. [3]



Q 26. State Gauss law. Use it to derive electric field due to uniform sheet of charge [3]

- Q 27.** (a) Suppose that the electric field amplitude of an electromagnetic wave is $E_0 = 120 \text{ N/C}$ and that its frequency is $\nu = 50.0 \text{ MHz}$. [3]
- Determine, B_0 , ω , k , and λ .
 - Find expressions for E and B .

OR

- (b) A parallel plate capacitor made of circular plates each of radius $R = 6.0 \text{ cm}$ has a capacitance $C = 100 \text{ pF}$. The capacitor is connected to a 230 V ac supply with a (angular) frequency of 300 rad s^{-1} . [3]
- What is the rms value of the conduction current?
 - Is the conduction current equal to the displacement current?
 - Determine the amplitude of B at a point 3.0 cm from the axis between the plates.

- Q 28.** (a) A circular ring of diameter 0.2 m is placed in a uniform magnetic field of 0.4 T . The ring is rotated about its diameter at a frequency of 60 Hz . [3]
- If the ring has 50 turns, then what is the maximum induced emf in the ring?
 - State one condition under which the induced emf in the circular ring will be zero?

OR

- (b) A solenoid has a core of a material with relative permeability 400. The winding of solenoid is insulated from the core and carry a current of 2 A . If the number of turn is 1000 per metre, calculate [3]
- H
 - M
 - B

Section D - Long Answer Questions

- Q 29.** (a) i. When refraction occurs of a convex spherical refracting surface and the ray travels from rarer to denser medium establish relation between image distance, object distance and radius of curvature of refracting surface. [5]
- ii. Using the above expression derive lens makers formula.

OR

- (b) i. Derive expression for angular magnification produced by refraction type telescope in normal adjustment using proper diagram [5]
- ii. Draw the diagram of reflecting type telescope. Give three reasons why reflection type telescope is superior to that of refraction type telescope.

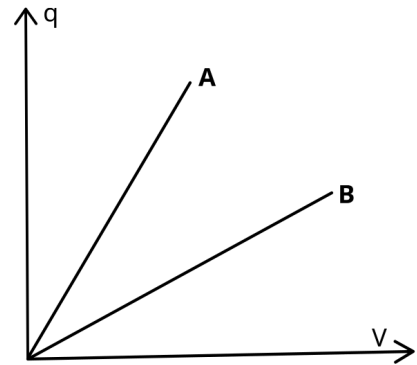
- Q 30.** (a) i. A camera usually operates at 1.5 V and this potential difference is not sufficient to emit light energy using flash. For this purpose, the flash circuit of the camera has a capacitor that is charged to 300 V - 330 V using various electrical components. If the voltage generated across the plates of the capacitor is 300 V and the capacitance of the parallel plate capacitor used is $100 \mu\text{F}$, then find the energy released when the trigger button on the camera is pressed. [5]

- ii. How much charge does the **100 μF** capacitor charged to **300 V** hold?
- iii. If the distance between the parallel plate capacitor of capacitance **100 μF** is increased two times, then calculate the capacitance of the capacitor.
- iv. The graph below shows the variation of charge '**q**' with potential difference '**V**' for a parallel plate capacitor '**C**' for scenarios P and Q.

Scenario P - the space between the capacitor '**C**' is filled with **air**.

Scenario Q - the space between the capacitor '**C**' is filled with a substance of **dielectric** constant **K**.

Which of the two lines **A** or **B** corresponds to scenario **Q**? Give a reason for your answer.

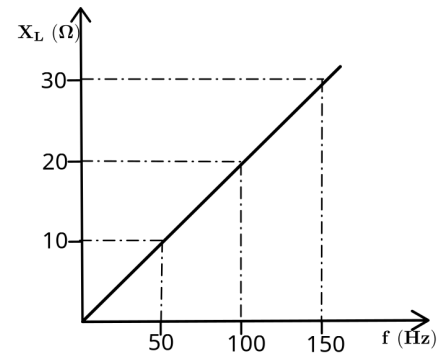


OR

- (b) i. Define the capacitance of a capacitor. Obtain the expression for the capacitance of a parallel plate capacitor in terms of plate area A and separation d between the plates. [5]
- ii. A slab of material of dielectric constant K has the same area as the plates of a parallel plate capacitor but has a thickness $\frac{3d}{4}$. Find the ratio of capacitance with dielectric inside it to its capacitance without dielectric.

- Q 31.** (a) An inductor of inductance '**L**' is connected to an AC source, **$V = 100 \sin \omega t$** .

The graph below represents the variation of inductive reactance X_L of the inductor with the frequency of an alternating source.



- i. What is the self-inductance of the inductor?
- ii. If the ac source is replaced by a battery such that **$V = 100 \text{ V}$** , then what is the inductive reactance of the inductor? Give reason.
- iii. When the frequency is **50 Hz**, what is the average power dissipated by the inductor over a complete cycle in the circuit? Justify your answer.
- iv. This inductor is connected in series with a resistance of **15 Ω** and a capacitor of **5 μF** . The frequency of the alternating source is varied such that the power dissipated in the circuit becomes maximum. Calculate the frequency and the phase difference between alternating voltage and current when the power dissipated is the maximum.

OR

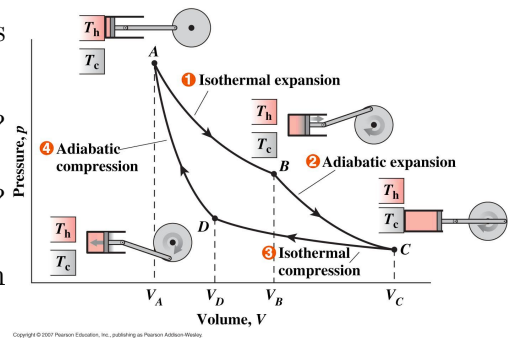
- (b) i. Describe the construction and working of a transformer and hence obtain the relation for $\left(\frac{V_s}{V_p}\right)$ in terms of number of turns of primary and secondary. [5]
- ii. Discuss four main causes of energy loss in a real transformer.

Section E - Case Study

Q 32. Variation of Photoelectric current with Potential Difference Figure shows the variation of photoelectric current measured in a photocell circuit as a function of the potential difference between the plates of the photocell when light beams A, B, C and D of different wavelengths are incident on the photocell. [4]

Examine the given figure Answer the following questions based on the above:

- Which light beam has the highest frequency and why?
- Which light beam has the longest wavelength and why?
- Which light beam ejects photoelectrons with maximum momentum and why?
- What is the effect on threshold frequency and stopping potential on increasing the frequency of incident beam of light? justify your answer.



Q 33. Before nineteenth century, scientists believed that magnetic properties were confined to a few materials like iron, cobalt and nickel. But in 1846, Curie and Faraday discovered that all the materials in the universe are magnetic to some extent. These magnetic substances are categorized into two groups. Weak magnetic materials are called diamagnetic and paramagnetic materials. Strong magnetic materials are called ferromagnetic materials. According to the modern theory of magnetism, the magnetic response of any material is due to circulating electrons in the atoms. Each such electron has a magnetic moment in a direction perpendicular to the plane of circulation. In magnetic materials all these magnetic moments due to the orbital and spin motion of all the electrons in any atom, vectorially add upto a resultant magnetic moment. The magnitude and direction of these resultant magnetic moment is responsible for the behaviour of the materials. For diamagnetic materials χ is small and negative and for paramagnetic materials χ is small and positive. Ferromagnetic materials have large χ and are characterized by non-linear relation between B and H . [4]

- The universal (or inherent) property among all substances is

A. Diamagnetism B. Paramagnetism C. Ferromagnetism D. Both (A) and (B)

- Which of the following is weakly repelled by a magnetic field

A. Iron B. Cobalt C. Steel D. Copper

- The value of the magnetic susceptibility for a super conductor is

A. Zero B. Infinity C. -1 D. +1

- Two wires of the same length are shaped into a square and a circle if they carry the same current, ratio of magnetic moments is

A. $2 : \pi$ B. $\pi : 2$ C. $\pi : 4$ D. $4 : \pi$